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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCPES100 -MOVEMENT EDUCATION

(2024 ADMISSION ONWARDS)

Answer Key

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Part A

MCQ

Answer all questions

Each question carries 1 mark

1. Which type of movement involves maintaining the body in a stationary position while controlling posture?

d). Static movement

2. Which of the following is a key element in Movement Education?

c). Body awareness

3. Which of the following best defines Movement Education?

b). The use of physical activities to promote learning and development

4. Which area of the brain is primarily responsible for the regulation of attention and focus?

c). Prefrontal cortex

5. Name the two nutrients that act as antioxidants supporting immune function.

c). Vitamin C & E

6. Which mineral is necessary for the production of thyroid hormone?

a). Iodine

7. develop the term learning by doing.

a). Gaining knowledge through practical experience.

8. which is the building blocks of cell growth and division?

b). Protein

9. Why is fun an essential aspect of movement education in early childhood?

b). It promotes children's emotional and social development

10. Which of the following is a key benefit of adapted physical activities for individuals with disabilities?

b). Improved social interaction

[1x10=10]

Part B

Short Answer

Answer any 10 questions

Each question carries 2 marks

11. What is the difference between gross and fine motor skills?

- Gross motor skills: Large muscle movements (running, jumping, walking)
- Fine motor skills: Small muscle movements (writing, drawing, buttoning)

12. How does physical activity help in strengthening the connection between attention and memory?

- Improves blood flow to brain
- Enhances concentration
- Strengthens neural connections
- Improves memory retention

13. Analyse how different movement patterns affect a child's spatial perception.

- Helps understand direction, distance, and position
- Improves coordination and awareness of surroundings
- Develops ability to judge space while moving

14. Define brain development and function in nutritional influence on growth and development.

- Brain development - growth of neurons and connections
- Nutrition provides energy and nutrients (like iodine, protein)
- Proper nutrition - better cognitive and motor development

15. How does the tonic neck reflex help infants in their early development?

- Helps develop coordination between head and limbs
- Prepares infants for voluntary movement

- Important for early motor development

16. How do fundamental movements in toddlerhood differ from more complex movements in preschoolers?

- Fundamental: basic skills (running, jumping)
- Complex: advanced skills combining basics (sports movements)
- Complex skills develop from fundamental skills

17. How protein synthesis helps tissue repair and maintenance

- Builds and repairs body tissues
- Essential for muscle growth
- Important for physical development

18. How can motion capture technology be used to improve a child's motor skills?

- Tracks body movements digitally
- Helps to analyse and improve motor skills
- Used in sports training and rehabilitation

19. Explain how problem-solving is encouraged through sports and games.

- Encourages decision-making
- Improves critical thinking
- Teaches strategy and teamwork

20. Mention two ways technology can be used to enhance physical activity in children's play.

- Interactive games (e.g., fitness apps)
- Motion tracking devices
- VR (Virtual reality)

21. What is the role of motion tracking in VR for child movement education?

- Creates realistic movement simulation
- Helps skill training and coordination
- Improves learning experience
- Helps in postural correct or deformities

22. Explain the significance of motor skill development in movement education for differently abled individuals.

- Improves independence
- Enhances physical strength
- Boosts confidence
- Supports social inclusion

[2x10=20]

Part C

Short Essay

Answer any 4 questions

Each question carries 5 marks

23. Discuss the importance of space awareness in movement education.

- Helps in understanding body position in relation to surroundings.
- Improves coordination and control of movements.
- Ensures safety by avoiding collisions and injuries.
- Develops concepts of direction, levels, and pathways.
- Enhances teamwork and cooperation in group activities.
- Encourages creativity in activities like dance and gymnastics.
- Supports better performance in sports and daily movements.

24. Explain how memory plays a role in performing complex movement patterns such as dance routines.

- Helps in learning and recalling movement sequences.
- Involves short-term and long-term memory for retention.
- Develops muscle memory (procedural memory) for automatic actions.
- Improves timing and rhythm in activities like dance.
- Enables smooth transitions between movements.
- Aids in error correction and improvement.
- Reduces mental effort, allowing focus on expression and coordination.

25. Discuss the impact of genetic disorders such as Turner syndrome and Klinefelter syndrome on growth and development.

- Both are genetic disorders affecting sex chromosomes.
- Turner syndrome causes short stature and delayed puberty in females.
- Klinefelter syndrome leads to tall stature and low muscle mass in males.
- Both may cause motor skill and coordination difficulties.
- Can result in learning and developmental challenges.
- Affect physical fitness and participation in activities.
- Early treatment and support can improve quality of life.

26. Describe how specialized activities like sports or dance help refine motor skills in young children

- Improve gross motor skills like running and jumping.
- Enhance fine motor control and coordination.
- Develop balance, flexibility, and agility.
- Promote practice and repetition for skill mastery.
- Improve rhythm and timing (especially in dance).
- Build confidence and social interaction.
- Support overall physical and mental development.

27. Discuss the impact of sensory deprivation on motor skill development in early childhood.

- Reduces sensory input needed for movement learning.
- Causes delay in motor skill development.
- Affects balance, coordination, and posture.
- Weakens hand-eye coordination and control.
- Leads to low confidence in movement.
- Impacts cognitive and emotional development.
- Requires early intervention and enriched environment.

28. How does AI contribute to adaptive learning in child movement education?

- Provides personalized learning based on individual ability.
- Uses motion tracking for real-time feedback and correction.
- Adjusts difficulty level automatically.
- Increases engagement through interactive tools and games.
- Supports children with special needs (inclusive learning).
- Helps teachers with data analysis and progress tracking.
- Improves efficiency and effectiveness of skill learning.

[5x4=20]